

Detecting Hate and Inflammatory Content in Bengali Memes: A New Multimodal Dataset and Co-Attention Framework

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Abstract—Internet memes have become a dominant form of expression on social media, including within the Bengali-speaking community. While often humorous, memes can also be exploited to spread offensive, harmful, and inflammatory content targeting individuals and groups. Detecting this type of content is exceptionally challenging due to its satirical, subtle, and culturally specific nature. This problem is magnified for low-resource languages like Bengali, as existing research predominantly focuses on high-resource languages. To address this critical research gap, we introduce Bn-HIB (Bangla Hate Inflammatory Benign), a novel dataset containing 3,247 manually annotated Bengali memes categorized as Benign, Hate, or Inflammatory. Significantly, Bn-HIB is the first dataset to distinguish inflammatory content from direct hate speech in Bengali memes. Furthermore, we propose the MCFM (Multi-Modal Co-Attention Fusion Model), a simple yet effective architecture that mutually analyzes both the visual and textual elements of a meme. MCFM employs a co-attention mechanism to identify and fuse the most critical features from each modality, leading to a more accurate classification. Our experiments show that MCFM significantly outperforms several state-of-the-art models on the Bn-HIB dataset, demonstrating its effectiveness in this nuanced task.

To facilitate reproducibility and future research, the Bn-HIB dataset has been made publicly available through Mendeley Data.

Warning: This work contains material that may be disturbing to some audience members. Viewer discretion is advised.

Index Terms—Bengali, Hate-Speech, Multi-Modal, Meme

I. INTRODUCTION

Image-text memes have become a powerful communication medium but also a vehicle for hate speech, misinformation, and polarization. Their multi-modal and culturally nuanced nature makes harmful intent detection difficult, especially in low-resource languages like Bengali, where code-mixing and symbolic visuals challenge existing English-centric moderation systems. As memes increasingly shape social interactions,

TABLE I: Annotation Inclusion And Exclusion Criteria

Class	Inclusion Criteria	Exclusion Criteria
Inflammatory	– Controversial statements or imagery; norm-challenging discourse; inflammatory tone or humor; boundary-pushing content.	– Accidental controversy; mild disagreement; primarily educational focus.
Hate	– Specific names or groups; group-identifiable traits; pronouns with clear referents; direct or indirect addressing.	– Generic mentions; self-referential content.
Benign	– Positive or neutral tone; everyday relatable humor; inclusive, non-divisive content.	– Any provocative or targeting cue; implicit controversial element.

research interest in meme analysis has grown substantially. Prior works explored various communicative dimensions such as sentiment and emotion recognition [1], sarcasm detection [2], and offensive content identification [3]. The rise of toxic memes has further driven studies on hate [4]–[6], offensiveness [7], [8], and harmful intent [9]. However, these efforts have largely focused on high-resource languages (e.g., English, Hindi), leaving low-resource languages underexplored. Only a few works, such as those on Tamil, Malayan, and other regional languages [10]–[12], have addressed objectionable memes, revealing a critical gap in low-resource meme moderation research. Bengali memes have recently gained wide popularity, shaping public opinion but often promoting negativity and violence. Prior studies have explored abusive [13], hateful [14]–[17], and aggressive [18] Bengali memes. However, these works are limited to binary or topic-specific hate detection (e.g., religion, gender, politics etc.).

Research in low-resource settings has largely overlooked the distinction between hateful and inflammatory memes. While not all hateful memes are designed to provoke communal tensions or incite violence, inflammatory memes subtly manip-



Fig. 1: Meme (a) portrays Hindus as hypocrites for allegedly destroying their own temples and blaming Muslims, while Meme (b) specifically targets an organization and individual, making it more directed but less representative of the broader Hindu population.

ulate public perception, fostering generalized negative attitudes toward specific groups. Figure 1 illustrates this distinction: although Fig. 1a may appear hateful, a detailed analysis of its textual and visual context classifies it as inflammatory due to its implicit manipulative persuasive intent. Previous systems fail to differentiate such subtle cases, highlighting the need for dedicated approaches.

Motivated by this gap, we introduce a novel corpus of Bengali memes, differentiating inflammatory, hateful, and benign content. On the technical front, state-of-the-art multimodal systems struggle with meme analysis, particularly when subtlety or implicit meaning is involved. Memes often exhibit weak alignment between text and visuals and contain significant noise compared to structured multi-modal data. To address these challenges, we propose a Multi-Modal Co-Attention Fusion Model (MCFM).

- We develop a novel multi-modal dataset Bn-HIB consisting of 3,247 Bengali memes labeled as Hate, Benign and Inflammatory.
- We propose MCFM, a simple yet effective multimodal fusion model that leverages both combined and modality-specific features to capture subtle elements.
- Finally, We conduct extensive experiments showing that MCFM outperforms fifteen state-of-the-art unimodal and multimodal baselines across all evaluation metrics.

II. RELATED WORK

This section reviews prior studies on multimodal hate speech detection datasets in both high-resource and low-resource language settings.

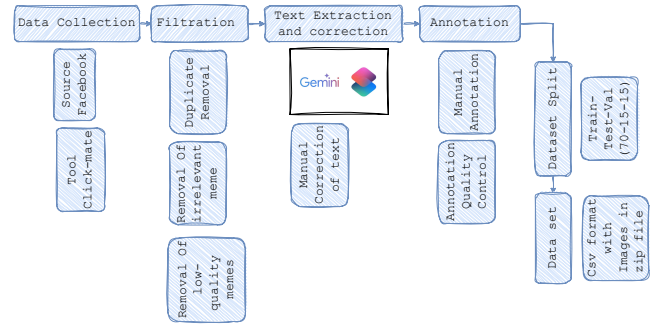


Fig. 2: Data Collection and Pre-Processing Overview

Multi-Modal Hate Speech Detection Datasets In High-Resource Settings: Several studies have explored multimodal information for hate speech and offensive content detection in recent years. [19] developed a multimodal dataset for offensive meme detection. Similarly, [20] and [21] introduced multimodal datasets focused on detecting hateful content from online memes. [9] proposed the MOMENTA dataset for harmful meme detection, incorporating both textual and visual modalities. In addition, two multimodal datasets have been developed for the Hindi language: one by [11] and another by [22], both aimed at identifying offensive and hateful memes in a high-resource context.

Multi-Modal Hate Speech Detection Datasets In Low Resource Settings: Research on multimodal hate speech detection in low-resource languages such as Bengali has gained attention more recently. [23], [16], [15], and [17] introduced multimodal hate speech datasets addressing the Bangla language. [13] developed a dataset for abusive meme detection in Bengali, while [18] focused on aggression detection in similar multimodal contexts. Researchers have employed various fusion techniques to combine visual and textual modalities, including early and late fusion methods, and the use of pretrained vision-language models such as CLIP [16], [18], [24], BLIP [18], ALBEF [16], [18], and MMBT [16]. Some works have also introduced custom architectures [16], [18] tailored to their specific tasks.

Differences From Existing Research: While significant progress has been made in multimodal hate speech and offensive content detection, there remains a notable gap in the study of inflammatory meme detection, particularly for low-resource languages such as Bengali. To the best of our knowledge, no prior work distinguishes hate from inflammatory content within this domain. Moreover, most existing datasets focus on binary classifications (e.g., hateful vs. non-hateful), limiting their capacity to capture nuanced or targeted offensive content—such as political, gendered, or religious themes—which are prevalent and rapidly evolving across social media platforms. Most of the existing datasets are entirely in Bengali [13], [15], [18], [23], [24]. Only one study focuses on Bengali and code-mixed Bengali [16], while another includes code-mixed, code-switched, and Bengali Memes [17]. Furthermore,

No previous study has publicly disclosed the specific sources of their meme collections.

In light of these identified gaps, our work differs from the existing works in three significant ways: (i) we develop a dataset distinguishing *hate*, *inflammatory*, and *benign* memes in Bengali, Bengali-English code-mixed, and code-switched contexts; (ii) we provide an annotation guideline to support dataset creation for other low-resource languages. (iii) we identified and shared Public Bengali Facebook meme pages and groups with front-page screenshots for efficient meme sourcing for future researchers.

III. DATASET

Description of Classes:

- **Hate Memes (HM):** Explicitly hostile content targeting individuals or groups through derogatory language, visuals, or stereotypes, aiming to incite hatred.
- **Inflammatory Memes (IM):** Implicitly Inflammatory content using sarcasm, irony, or coded cues to incite polarization or unrest without overt hate speech.
- **Benign Memes (BM):** Non-hostile, humorous, or cultural content intended for entertainment or social bonding without divisive intent.

Data was collected from 25 public Facebook groups/pages known for high meme activity using manual curation followed by the Click-mate [25] macro tool for scalable extraction. Only publicly available content was used, yielding approximately 5,000 raw memes. After filtering for multi-modality and quality as [15], 3,247 memes remained. Text embedded in images was extracted using the Gemini API [26], manually verified for Bengali script accuracy, and paired with corresponding images.

Annotation followed open coding of 1,000 randomly sampled memes, revealing 15 categories grouped into the three overarching categories above. Three trained annotators—fluent in Bengali and code-mixed dialects—used Label Studio [27] for labeling, viewing both image and OCR text. After a pilot phase on 300 memes (initial Fleiss’ $\kappa = 0.65$), iterative calibration raised agreement to 0.79. Disagreements were resolved through expert arbitration considering irony, symbolism, and contextual nuance.

The final dataset comprises 3,247 annotated memes, divided into training (70%), validation (15%), and test (15%) splits. Table II presents dataset statistics; Figs. 2 and 4 show the data collection and pre-processing workflow and decision rules for annotators. The dataset adheres to ethical, culturally sensitive, and bias-minimized annotation principles as [18]. Annotation inclusion and exclusion criteria are summarized in Tab. I. Figure 7 illustrates examples from each category.

IV. METHODS

A. Models And Fusion Strategies

Table III summarizes the textual, visual, and multi-modal fusion strategies used. Textual encoders capture linguistic features in code-mixed or multilingual text, while visual encoders extract semantic and spatial features from meme images. Multi-modal fusion strategies—including early/late

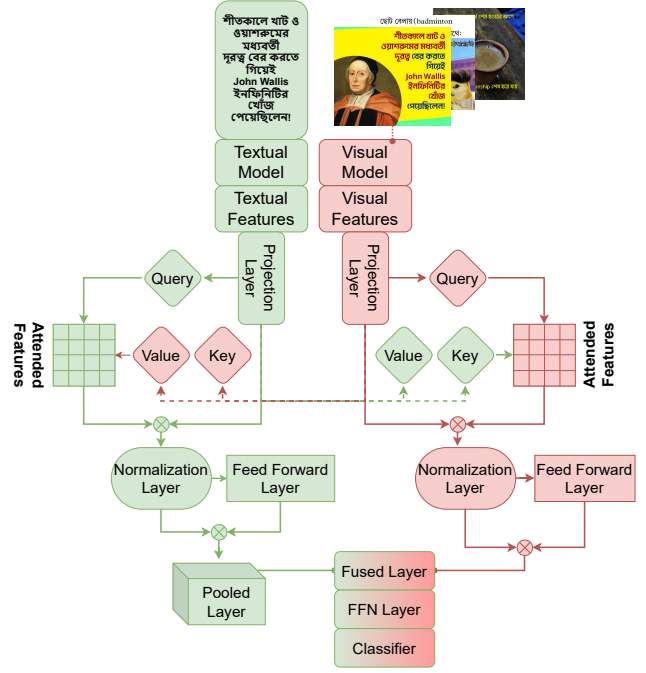


Fig. 3: Proposed Multi-Modal Co-Attention Fusion Model (MCFM)

TABLE II: Dataset Statistics

Category	Class Split			Language Split			Text Length (Words)			
	Train	Val	Test	Train	Val	Test	Mean	Max	Min	Mode / Med.
Hate (HM)	811	174	173	—	—	—	—	—	—	—
Inflammatory (IM)	773	166	167	—	—	—	—	—	—	—
Benign (BM)	688	147	148	—	—	—	—	—	—	—
Bengali	—	—	—	1190	255	256	—	—	—	—
Code-switched	—	—	—	332	72	71	—	—	—	—
Code-mixed	—	—	—	750	160	161	—	—	—	—
Overall							121	15089	4	60 / 90

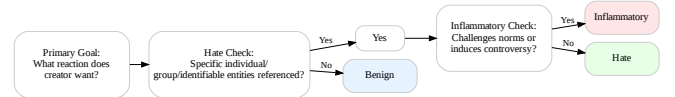


Fig. 4: Annotation Decision Tree

TABLE III: Textual, Visual, and Multi-Modal fusion Models

Category	Model / Strategy	Ref.
Textual	Banglish-BERT	[28]
	mBERT	[29]
	XLM-RoBERTa	[30]
	MuRIL	[31]
	mDeBERTa-v3	[32], [33]
Visual	ResNet-50	[34]
	ConvNeXt	[35]
	ViT-B/16	[36], [37]
	Swin V2	[38]
	EfficientNet-B3	[39]
Fusion	FocalNet-Base-LRF	[40]
	Early Fusion	
	Late Fusion	
	CLIP	[41]
	Cross-Attention	[42]
	MCFM(ours)	

TABLE IV: Model Performance (Accuracy / Macro F1)

Textual Models			Visual Models		
Model	Acc.	F1	Model	Acc.	F1
Banglish BERT	0.7070	0.7109	ResNet-50	0.5553	0.5509
XLM-R	0.7152	0.7162	EffNet-B3	0.5205	0.5205
M-BERT	0.6865	0.6906	ViT-B16	0.4734	0.4790
Muril	0.6475	0.6462	ConvNext	0.5738	0.5731
<i>MDeBERTa-V3</i>	0.7193	0.7235	<i>Swin-V2</i>	0.5922	0.5909
FocalNet-lrf	0.5512	0.5506			
Multimodal Models					
Late Fusion	0.6230	0.6201	CLIP	0.7295	0.7322
Early Fusion	0.6373	0.6440	MCFM	0.7746	0.7765
Cross-Att (T→I)	0.7643	0.7649	Cross-Att (I→T)	0.7684	0.7718
Improvement vs MDeBERTa-V3: +0.0553 Acc., +0.0530 F1					

fusion, CLIP [41]-based integration, and attention-driven fusion enhance joint understanding of text and images. Fig 3 shows the architecture of our proposed model.

B. Experimental Details

To ensure reproducibility, the experimental details for all models—textual, visual, and multimodal—are consolidated in Table V. These specifications cover architectural variants, hyperparameters, and training procedures.

C. Evaluation Metrics

We evaluate performance on 3247 instances across three classes using Accuracy and Macro F1-score, prioritizing the latter due to minor class imbalance.

Accuracy measures the proportion of correct predictions:

$$\text{Accuracy} = \frac{1}{3247} \sum_{i=1}^{3247} \mathbb{I}(y_i = \hat{y}_i), \quad (1)$$

where y_i and $\hat{y}_i$ are the true and predicted labels, and $\mathbb{I}(\cdot)$ is the indicator function.

Macro F1-score is the unweighted average of per-class F1-scores:

$$\text{F1}_{\text{macro}} = \frac{1}{3} \sum_{k=1}^3 \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}, \quad (2)$$

with $\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}$ and $\text{Recall}_k = \frac{TP_k}{TP_k + FN_k}$, where TP_k , FP_k , and FN_k are true positives, false positives, and false negatives for class k .

V. RESULTS

Independent textual and visual classifiers show Table IV that MDeBERTa-V3 [32], [33] achieved the highest textual Macro F1 (0.7235), followed by XLM-RoBERTa [30] (0.7162) and Banglish- BERT [28] (0.7109), benefiting from disentangled attention and multilingual masked decoding. Visual-only models lagged: Swin Transformer-V2 [38] (0.5909), ConvNeXt [35] (0.5731), ResNet-50 [34] (0.5509), reflecting low-res and blurry image challenge. Our fusion models leverage the best-performing unimodal models. Fusion models improved performance Table IV: Early Fusion (0.6440) slightly outperformed Late Fusion, CLIP [41] exceeded textual baselines (0.7322),

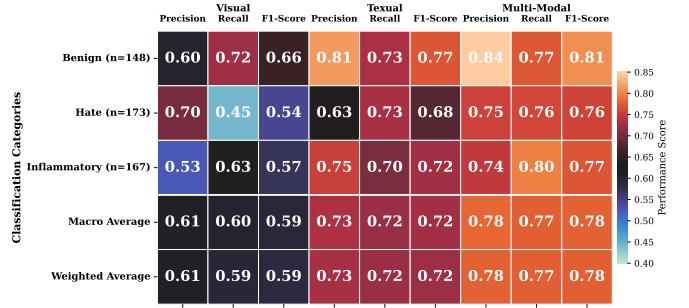


Fig. 5: Class-Wise Performance Of Top Models

and our MCFM topped all (0.7765) via bidirectional modality alignment; Cross-Attention [42] Image→Text (0.7718) indicates visual context aids semantic alignment. Multimodal models also reduced class-wise variance (see Fig 5).

A. Error Analysis

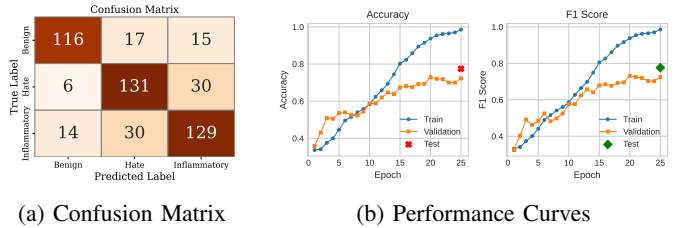


Fig. 6: Error analysis: confusion matrix and performance trends.

As shown in Fig. 6b, the model overfit on the validation set despite fair test generalization. Fig. 6a shows that Benign memes had the fewest errors (32: 17 hate, 15 Inflammatory), while Inflammatory memes had the most (44: 30 hate, 14 Benign). The hate class was often confused with Inflammatory (30 cases), reflecting semantic overlap. High-confidence (>80%) errors were dominated by the hate class (53.84%), whereas low-confidence (<50%) correct predictions indicated ambiguity between hate and Inflammatory categories. Frequent errors stemmed from (i) conflicting text–image cues, (ii) unrecognized satire, sarcasm, or cultural references, and (iii) linguistic noise such as Bengali neologisms, OCR artifacts, and code-mixed text. Limited visual detail (e.g., small logos or symbols) further degraded accuracy. Overall, these results highlight key multimodal moderation challenges—implicit humor, cultural nuance, and fine-grained semantic fusion—suggesting future focus on improved contextual and cross-modal understanding.

VI. CONCLUSION

This paper presented Bn-HIB, a novel multimodal dataset of 3,247 memes across three classes, and proposed the Multimodal Co-Attention Fusion Model (MCFM). Experiments showed that MCFM surpasses fifteen state-of-the-art unimodal and multimodal baselines, validating its effectiveness. Future directions include expanding Bn-HIB to more domains

TABLE V: Experimental Details of Textual, Visual and Multi-Modal Models

Component	Textual (MDeBERTa-V3)	Visual (Swin-B)	Multi-Modal (Fusion)
Base Model	microsoft/mdeberta-v3-base	Swin-B (IMAGENET1K_V1)	Transformer + ImageNet-pretrained CNN
Hidden Size	768	—	Text: 1024
Input Size	Max token length = 256	224 × 224	224 × 224
Tokenization / Preprocessing	Native tokenizer	Flip, rotation, jitter (train); resize + normalize (eval)	Tokenizer + image preprocessing
Special Tokens	[CLS], [SEP], [PAD]	—	—
Classification Head	ReLU → Dropout(0.5) → Linear (3 classes)	Dropout(0.5) → Linear (3 classes)	Dropout(0.5) → Linear
Fusion Architecture	—	—	Co-attention (MCFM), Cross-attention, CLIP-style
Fusion MLP	—	—	2-layer MLP with GELU
Attention Heads	—	—	8
Optimizer	AdamW	Adam	AdamW
Learning Rate	1×10^{-5}	1×10^{-4}	2×10^{-5}
Weight Decay	0.01	—	0.01
Batch Size	8	8	Effective 8 (grad acc = 2)
Max Epochs	500	200	—
Early Stopping	Patience = 3 epochs	Patience = 3 epochs	Patience = 5 epochs (F1-based)
Loss Function	Cross-EntropyLoss	Cross-EntropyLoss	Focal Loss ($\alpha = 1.0$, $\gamma = 2.0$), Label Smoothing ($\epsilon = 0.1$)
Class Balancing	—	—	Inverse frequency weighting
Gradient Clipping	Max norm = 1.0	—	Max norm = 1.0
Learning Schedule	Linear warmup (10%)	—	Linear warmup (10%)
Precision	Mixed (FP16/FP32)	AMP	Mixed Precision (FP16)
Gradient Accumulation	—	—	Steps = 2
Efficiency Features	—	—	Gradient checkpointing, AMP
Regularization	Dropout, L2 decay	Dropout	Dropout, Label smoothing
Reproducibility	Seed 42, deterministic flags	Same	Same
Framework	PyTorch 1.12+, NVIDIA T4	PyTorch 1.12+, NVIDIA T4	PyTorch 1.12 + Transformers 4.20, NVIDIA T4



Fig. 7: Examples Of Classes

and languages, enabling fine-grained categorization for richer analysis, Develop more lightweight version of MCFM and integrating XAI to foster transparency, accountability, and user trust in content moderation.

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